

market in a big city. Most of them are usually condemned as military surplus but still work. Most Ham radio operators have such high-impedance headphones. If high-impedance headphones are not available, you can use the earphones of an old type of landline telephone receiver (dial type), or the handset of the latest pushbutton landline telephone.

For using the telephone handset for crystal radio, cut the wire that connects



Fig. 7: LM386 amplifier module



Fig. 8: Author's prototype in a box with speaker and battery

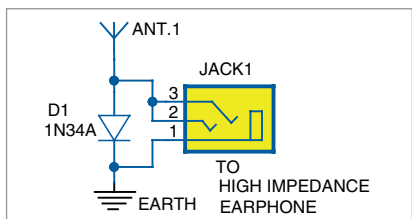


Fig. 9: Circuit 1 connections

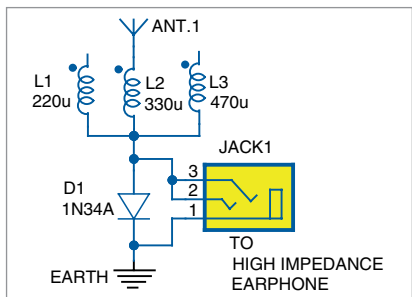


Fig. 10: Circuit 2 connections

the handset with the main unit. You will see four wires of different colours, of which two are connected to earphone and two to microphone. Ignore the microphone wires. Check the DC resistance across the two wires of the earphone/headphone using a multimeter. For best matching, it should be 300 ohms or above for use in crystal radios. In our experiment, earphone from pushbutton telephone handset with 150 ohms DC resistance also worked, but with less volume compared to high-impedance earphones. Connect a pair of miniature crocodile clips to the earphone wires, which will facilitate easy connection with the crystal radio.

If you are not able to get a high-impedance headphone or landline telephone's handset, you can use currently available low-impedance earphone of an MP3 player or cellphone (16 to 32 ohms impedance) along with an impedance matching transformer. The commonly available step-down transformer of 220V primary to 9V or 12V, 500mA secondary, which is used for low-voltage power supplies, can be used as an impedance-matching transformer. Everyone has a cellphone's earphone, which is also freely available in the market. Select an earphone with a sensitivity of 100 to 110dB/mW. Try different brands of earphones for the best results. Connect the primary winding of the matching transformer, as in the circuit, and connect the earphone to the secondary of the transformer. You will be able to listen to audio of the nearby MW station.

If you are getting a low volume in your headphone or earphone, you can always use a matching ampli-

fier and connect it in the place of the headphone. A mini amplifier based on LM386 IC is the best match for the crystal radios. It is low in cost and works off 3V to 9V battery. You can get a fully assembled LM386 amplifier module (refer Fig. 7) from an electronics market or assemble it yourself. Authors' prototype is shown in Fig. 8.

## Circuit 1

This circuit (refer Fig. 9) uses just two parts, one germanium diode and a pair or single high-impedance headphone, apart from an antenna and earth, as described earlier. The germanium diode is connected across the headphone terminals. The cathode of the diode is connected to the long wire antenna and anode to a good earth (metallic water pipe or the electrical socket). The polarity of the diode is not critical, and you can connect the diode in any direction as we use only half of the rectified signal.

Enjoy listening to the local station with high or low volume depending upon the length and height of the antenna, power of your local radio station, and how far away you are located from the transmitter. If you do not get sufficient volume with this circuit, you can try the second circuit with the addition of moulded inductors.

## Circuit 2

This circuit (refer Fig. 10) is similar to the first circuit, but a moulded inductor of 220, 330, or 470 micro-Henry is added in series between the antenna and the diode. Select any one of the inductors that gives better volume. As the inductor helps to select the MW band, a slight increase in audio output is observed.

## Circuit 3

In case you are not able to get the high-impedance headphone or pushbutton phone earphone, you can use your cellphone's or MP3 player's earphone having a low impedance of 16 to 32 ohms along with the impedance-matching transformer described earlier. As shown in Fig. 11, connect primary of the transformer to the terminals of the diode and secondary to the earphone. For best results,

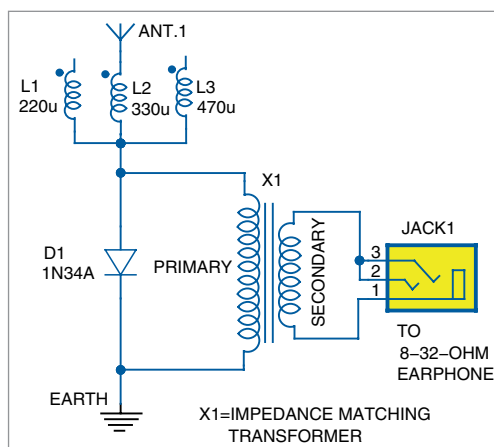


Fig. 11: Circuit 3 connections